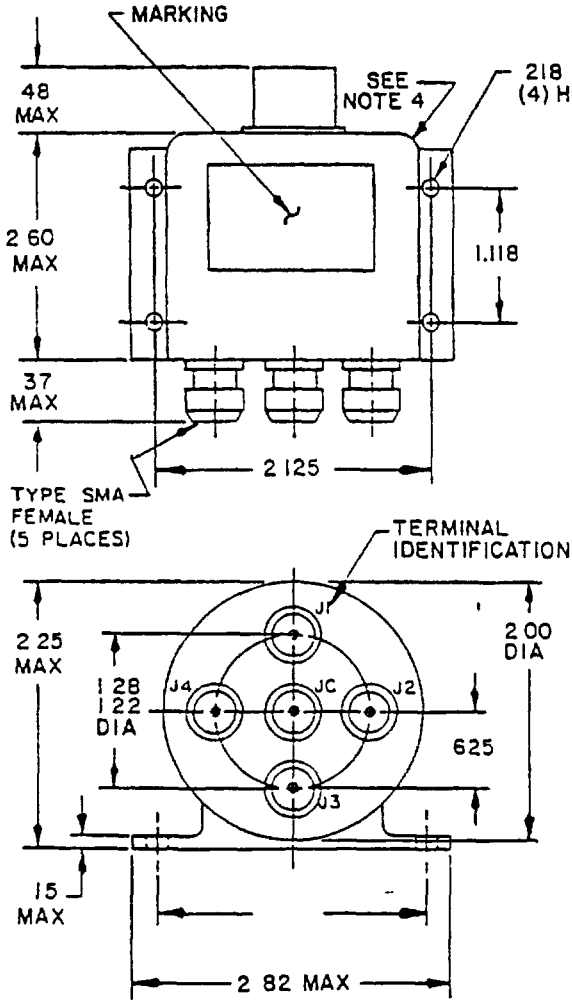
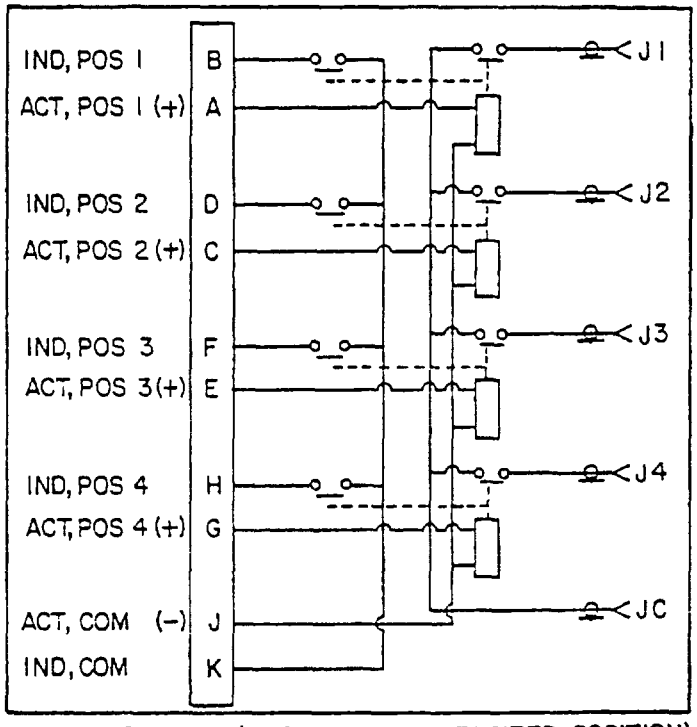
MILITARY SPECIFICATION SHEET

Ⓐ SWITCHES, RADIO FREQUENCY
TRANSMISSION LINE (COAXIAL) (ELECTRICALLY OPERATED)
CLASS 5, 1P4T

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-3928.





SCHEMATIC (SHOWN IN DEENERGIZED POSITION)

Inches	mm	Inches	mm
.15	3.8	1.28	32.5
.218	5.54	2.00	50.8
.37	9.4	2.125	53.98
.48	12.2	2.25	57.2
.625	15.88	2.60	66.0
1.118	28.40	2.82	76.6
1.22	31.0		

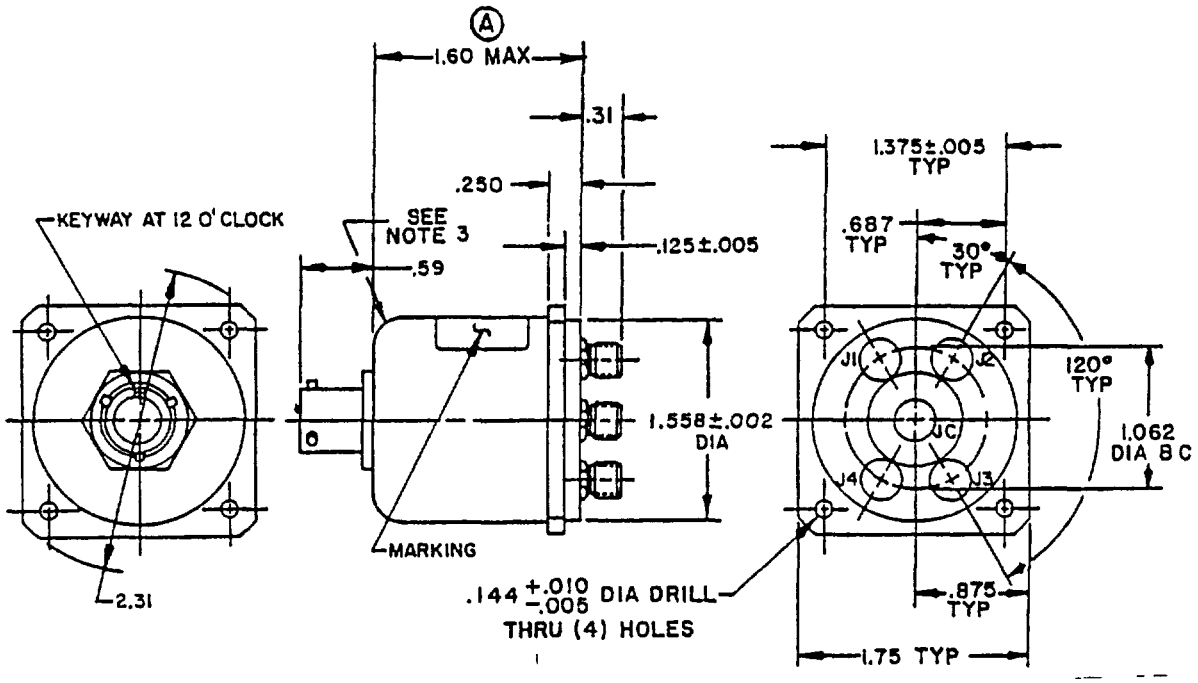
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place place decimals.
4. Round corners of case may be squared.

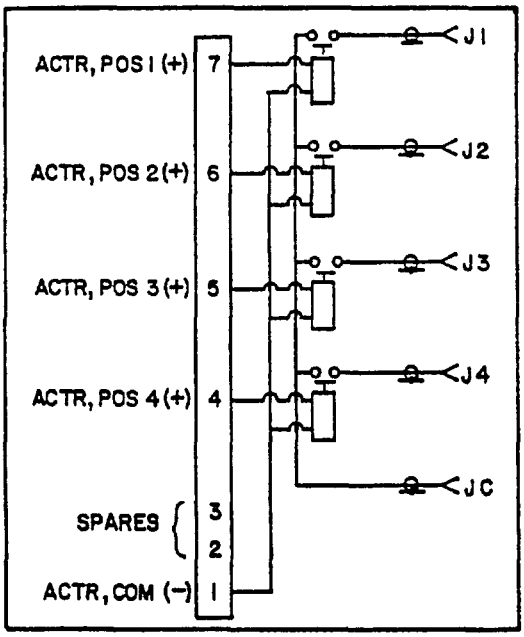
FIGURE 1. Switch configuration and schematic, part number M3928/17-01.

Ⓐ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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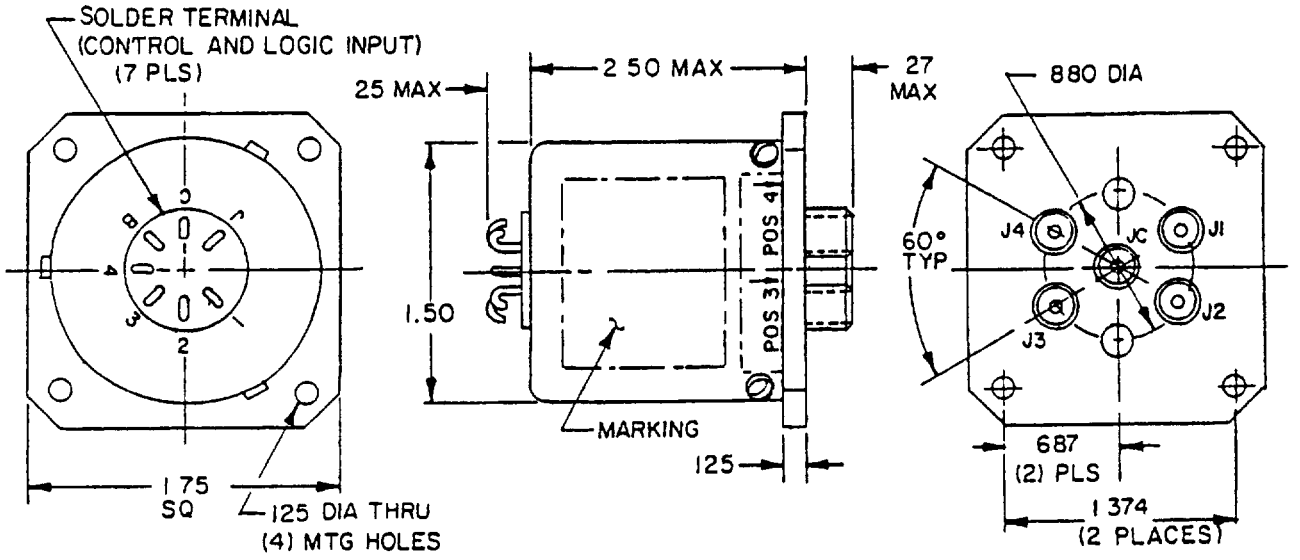
SCHEMATIC (SHOWN IN DEENERGIZED POSITION)

Inches	mm	Inches	mm	Inches	mm
.002	0.05	.250	6.35	1.375	34.93
.005	0.12	.31	7.9	1.558	39.57
.010	0.25	.59	15.0	1.60	40.6
.125	3.18	.687	17.45	1.75	44.5
.144	3.66	.875	22.23	2.31	58.7
		1.062	26.97		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±0.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

FIGURE 2. Switch configuration and schematic, part number M3928/17-02.

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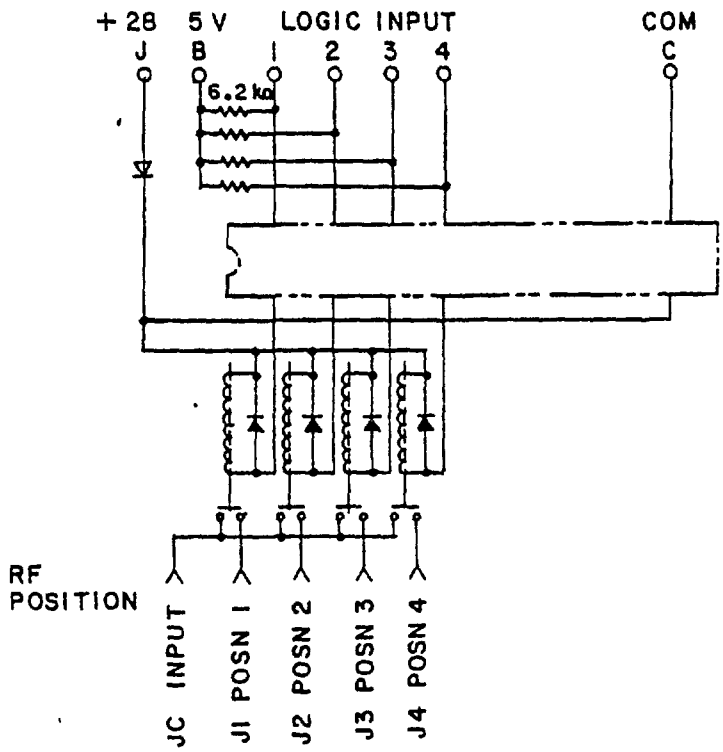


Inches	mm
.125	3.18
.25	6.4
.250	6.35
.27	6.85
.687	17.44
.880	22.35
1.50	38.1
1.75	44.5

NOTES.
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±0.03 (0.8 mm) for two place decimals.
4. Round corners of case may be squared.

Ⓐ FIGURE 3. Switch configuration and schematic, part number M3928/17-03.

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SCHEMATIC

Truth table/pin assignment:

Logic input terminals				RF positions			
1	2	3	4	1	2	3	4
1	0	0	0	On	-	-	-
0	1	0	0	-	On	-	-
0	0	1	0	-	-	On	-
0	0	0	1	-	-	-	On

FIGURE 3. Switch configuration and schematic, part number M3928/17-03 - Continued.

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(A) TABLE I Electrical and performance characteristics

Part no M3928/17-	Figure no	Housing	Frequency range dc to GHz	VSWR	Insertion loss (dB)	Isolation (dB)	Switching time (ms)	Position indication circuit and rating	Life cycles x 1000	Fail safe or latching	Operating current (A) 1/	Holding current (A) 1/	Nominal operating voltage	Pickup voltage (less than)	Dropout voltage (less than)	Power and indicator connector	Weight (oz)
				Max	Max	Min	Max				Max	Max					Max
2/ 01N,S	1	1	18	1 5 1	0 5	40	20	2A40 V dc	100	---	13	13	28 V dc	20 V dc	16 V dc	M83723/11W1210W	16
02N,S	2	1	18	1 5 1	0 5	60	20	None	1000	---	.17	17	28 V dc	20 V dc	16 V dc	RTK07 87P (Deutsch) or equal	16
3/ 03N,S	3	1	18	1 25 1 @ DC-6 GHz 1 50 1 @ 12-18 GHz	0 2 @ DC-6 GHz 0 5 @ 12-18 GHz	70 @ DC 6 GHz 60 @ 12 18 GHz	20	None	100	---	16	16	28 V dc +5 V dc 4/	20 V dc	16 V dc		5

1/ At 28 V dc and 20°C

2/ Transient interference (RFI)

3/ Logic voltage

Logic 0 0 to 0 8 V dc
Logic 1 2 4 V dc to 5 V dc

4/ +5 voltage deviation ± 5 V dc

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REQUIREMENTS:

Dimensions and configurations: See figures 1, 2, and 3.

RF connectors: Female connectors (5 places) shall meet the requirements of MIL-C-39012 and shall mate with SMA type male connectors per MIL-C-39012/55.

Electrical and performance characteristics: See table I.

(A) RF power handling capability (average): 25 watts (minimum) except dash number 03 is 15 watts (minimum).

Nominal impedance: 50 ohms.

Termination: Open.

Vibration: Method I.

Operating temperature: -55°C to +85°C.

Part number: M3928/17- (and dash number from table I).

Substitutability: Switches covered by this specification sheet are substitutable for the manufacturer's or original equipment manufacturer's (OEM) part numbers specified in table II. This information in no way implies that the manufacturer's or OEM part number or NSN is suitable as a substitute for the military part number.

TABLE II. Substitution data.

Part number M3928/17-	OEM or manufacturer's code	OEM or manufacturer's part number	NSN 5985-
01	07397	20-811082-1	00-007-5523
01	24022	CS18014-3	00-007-5523
01	82152	134D70200	---
02	82152	144C70100	01-042-0669
03	76301	5M2550-001	---
03	24022	CS38S14-T	01-160-0142

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5985-1016-9)

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